MILITARY SPECIFICATION

TEST SET, BATTERY

AN/AMM-1

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope. The equipment covered by this specification shall be capable of functioning as a d-c power source for Radiosonde Set AN/AMT-11( ), as a resistive load for activation of Battery BA-353/AM, and as a current and voltage monitor of the d-c power required for operation of Radiosonde Set AN/AMT-11( ).

1.2 Classification. The Test Set, Battery AN/AMM-1 shall be of one type and shall consist of the following:

Unit	Type Designation	Maximum Allowable Size (Inches) W	H	D	L	Maximum Allowable Weight (Pounds)	Applicable Paragraph
Test Set, Battery	TS-1101/AM	9	7-1/2		12	19	3.5
Cable, Power	W-1				48	0.5	3.5
Cable, Radiosonde	W-2				39	0.5	3.5

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

FSC 6625

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SPECIFICATIONS

Military

MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of
MIL-T-18303	Test Procedures, Preproduction, Acceptance, and Life, for Aircraft Electronic Equipment, Format for
MIL-S-19500	Semiconductor Devices, General Specification for
MIL-T-21200	Test Equipment for Use with Electronic and Fire Control Systems, General Specification for

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of U.S Military Property
MIL-STD-454	Standard General Requirements for Electronic Equipment

DRAWINGS

Naval Air Systems Command

57A119ASK1	List of Drawings for Test Set, Battery, AN/AMM-1
57A119D22	Panel Mounting Assembly

(Copies of specifications, standards drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Precedence. When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract. The contract shall have precedence over any specification.

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(2) This specification. This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be approved in writing by the Naval Air Systems Command.

(3) Referenced specifications. Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

3. REQUIREMENTS

3.1 First article. The battery test set furnished under this specification shall be a product which has been tested and has passed the first article tests specified herein.

3.2 Parts and materials. In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(a) Parts and materials as approved by MIL-T-21200 shall be given first consideration.

(b) When the characteristics of standard MIL parts and materials are such that their use will not fulfill the requirements because of size, weight, performance, or other reasons, material and parts shall be used which meet or exceed the requirements of the respective standard parts and material specification.

(c) When a nonstandard part is used, the meeting of general requirements such as nonflammable, etc., and the meeting of environmental conditions as required by MIL-T-21200 shall be of prime consideration.

3.2.1 Nonstandard parts and material approval. Approval for the use of nonstandard parts and materials shall be obtained as outlined in MIL-T-21200 except as follows:

(a) Mechanical parts shall be submitted as nonstandard parts only when the material requirements of MIL-T-21200 are not met.

(b) Nonstandard parts of an electrical nature as defined in MIL-T-21200 shall be approved before the first article model of the equipment is submitted for tests. The

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request for approval shall be forwarded to the Naval Air Systems Command before the part is purchased or fabricated. In addition to or in lieu of the Nonstandard Part Data Sheet, the contractor shall submit an itemized list of the parts, their sources, and the contractor's drawing or purchase specification covering the parts.

3.2.2 Diode types. The choice and application of electron diodes used in the equipment shall be in accordance with the following requirements and in the sequence noted:

(a) MIL-S-19500

(b) The use of any diode not meeting the requirements of (a) above shall be approved by the Naval Air Systems Command however, before consideration for approval, the contractor shall submit the following information:

(1) Quantitative data on the electrical advantages and mechanical adequacy of the proposed diode over the equivalent type on the acceptable listings.

(2) Test data and performance limits of the diode which must be maintained to ensure satisfactory and reliable performance in the equipment.

(3) Information as to whether the diode manufacturer has received Military or Navy qualification approval for the particular diode.

3.2.2.1 Diode complement A diode complement report shall be prepared and submitted to the Naval Air Systems Command for review and approval prior to first article testing. The diode complement shall include all diodes used in the equipment.

3.2.3 Performance objectives. Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in MIL-T-21200, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the Naval Air Systems Command for consideration. Each request shall be accompanied by complete supporting information.

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3.3 Design and construction:

3.3.1 General. This equipment shall fulfill all applicable design and construction requirements of MIL-T-21200 unless otherwise specified herein.

3.3.2 Service conditions. The equipment shall operate satisfactorily under the service (environmental) conditions contained in 3.3.2.1 through 3.3.2.4.

3.3.2.1 Temperature. As specified in MIL-T-21200 for Class 2 equipment.

3.3.2.2 Humidity. As specified in MIL-T-21200 for Class 2 equipment.

3.3.2.3 Vibration. The equipment shall not suffer damage or fail to give specified performance when subjected to the vibration test of 4.8.3.

3.3.2.4 Shock. As specified in MIL-T-21200 except that the impact shocks shall be 10g.

3.3.3 Standard conditions. The following conditions shall be used as a basis to establish normal performance requirements and for making laboratory bench tests on the equipment.

Temperature	Room ambient (30°C ± 10°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input Power	115 V AC ±5% 60 cycles 0.22 to 0.30 amperes

3.3.4 Total weight. The total weight of the equipment, including cables, shall be a minimum consistent with good design and shall not exceed 20 pounds.

3.3.5 Operational stability. The equipment shall operate with optimum performance for 200 hours, continuously or intermittently, without the necessity for readjustment of any of the controls which are not readily accessible on the front panel.

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3.3.6 Operating life.

3.3.6.1 Reliable operating life. The equipment shall have a reliable operating life of at least 200 hours without the necessity for bench servicing. Parts requiring servicing or replacement within this interval to renew this service life shall be specified by the manufacturer.

3.3.6.2 Total operating life The equipment shall have a minimum total operating life of 10,000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.3.7 Interconnection cabling. This equipment shall be capable of satisfactory operation with the cables which are furnished as part of the equipment. The cables shall be in accordance with the applicable requirements of MIL-T-21200. The external wiring shall be unshielded.

3.3.8 Cables and connectors. The equipment shall provide for the use of cables and connectors in accordance with MIL-T-21200.

3.3.9 Controls. All controls shall be satisfactory for mounting on the panel. These controls shall meet the requirements of MIL-T-21200.

3.3.10 Input power requirements. The equipment shall meet all applicable requirements and specified performances of MIL-T-21200. The power required shall not exceed the specified amounts.

A-C power (single-phase)

115-volt constant frequency

operating voltage limits	109 to 121 volts
operating frequency limits	55 to 65 Hertz
power (1Ø)	31 watts

3.3.11 Input power connection. Input power connections shall be in accordance with MIL-T-21200.

3.3.12 Overload protection. Overload protection for the internal power supply shall be provided in the equipment. All parts and circuits of the equipment which are likely to carry an overload due to any failures or poor adjustments shall have suitable protective devices or shall be proportioned to withstand such overload without permanent damage to the equipment. The use of fuses and other protective devices shall be held to a minimum.

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3.3.12.1 Under voltage protection. The equipment shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage returns to within the specified limits.

3.3.13 Nameplate. Nameplate approval for equipment identification shall be in accordance with NAVAIR Drawing Number 57A119D22.

3.3.13.1 Identification markings. The equipment identification markings, other than nameplates, shall be in accordance with MIL-STD-130.

3.3.14 Interchangeability. The equipment shall meet the interchangeability requirements as specified in MIL-T-21200. Unless otherwise specified, interchangeability shall be measured against a model supplied by the Naval Air Systems Command.

3.3.15 Interference control. The generation of interference by the equipment and the vulnerability of the equipment to interference shall be controlled within the limits of MIL-T-21200.

3.4 Performance.

3.4.1 Power source. The equipment shall operate from a 60-Hertz 115-volt a-c power source when the internal power supply is in use. For all other functions, the test unit shall be self-contained.

3.4.2 Warm-up time. The warm-up time for the equipment shall be a maximum of five minutes under normal (30°C) conditions and 15 minutes under extreme service conditions.

3.4.3 Dielectric strength. The equipment shall be capable of withstanding 500 volts d-c at room temperature and humidity for a period of 10 seconds when this voltage is connected between the chassis and each isolated circuit.

3.4.4 Insulation resistance. The circuits of the equipment shall have an insulation resistance of at least 10 megohms at 500 volts d-c when measured between the chassis and each isolated circuit.

3.4.5 Activation load. The load resistors of the activation loads shall be conservatively rated to hold temperature rise to a minimum during periods of continuous operation.

3.4.6 Electrical connections. All electrical connections shall be made in accordance with the applicable drawings listed in List of Drawings 57A119ASK1.

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3.4.7 Cables. The Power Cable W-1 and Radiosonde Cable W-2 shall be made in accordance with the applicable drawings listed in List of Drawings 57A119ASK1.

3.5 Detail requirements.

3.5.1 Battery Test Set TS-1101/AM, Power Cable W-1, and Radiosonde Cable W-2. The test set shall contain the parts and materials specified in the List of Drawings 57A119ASK1 and meet the following requirements:

3.5.1.1 Function. The test set shall provide:

(a) Power supply for Radiosonde Set AN/AMT-11( )

(b) Resistance loads for activating Battery, BA-353/AM

(c) Meters and switching circuits for monitoring the d-c operating power provided from either Battery BA-353/AM or the test set internal power supply during the period of battery activation and/or actual operation of the radiosonde set.

3.5.1.2 Case assembly. The Test Set, Battery TS-1101/AM shall be enclosed in a shallow rectangular metal case. The case shall include storage space for the instruction book and the two cables which are a part of the AN/AMM-1 system. The cover lid shall have attached to it a set of operating instructions. The sealed cover shall be held in a closed position by a spring loaded catch. With the cover in the open position and the cables connected, the unit shall permit operation through all controls mounted on the panel.

3.5.1.3 Form factor. The maximum dimensions of the case shall be as follows:

Length	H/D	Width
12 inches	7-1/2 inches	9 inches

3.5.1.4 Weight. The maximum weight of the equipment, including cables, shall be 20 pounds.

3.5.1.5 Connectors. The test set shall have the following connectors:

(a) Power connector. A three-pin male plug located on the upper part of the panel and to the right of center and marked POWER 115 V. 60~, shall provide connection to the external power source.

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(b) Radiosonde connector. A four-pin female plug located on the upper part of the panel and to the left of center and marked RADIOSONDE, shall provide connection to the Radiosonde Set AN/AMT-11( ).

(c) Battery connector. A cable located on the lower right section marked BATTERY CABLE, shall provide the battery connection to the test unit.

3.5.1.6 Meters. The test set shall have the following meters:

(a) "A" current-voltage meter. A one-milliampere meter, located in the upper left section of the panel, shall indicate A current and voltage. The meter shall normally indicate voltage but provide current indications through a momentary-contact, push-button switch. The current switch shall be located below the meter.

(b) "B" current-voltage meter. A one-milliampere meter located in the upper right section of the panel, shall indicate B current and voltage. The meter shall normally indicate voltage but provide current indications through a momentary-contact, push-button switch. The current switch shall be located below and to the left of center of the meter. In addition, C voltage indication shall be provided through a second push-button switch located below and to the right of center of the meter.

(c) Calibration. The calibration shall be confined to color shading of the meter scales to show areas of acceptable performance. These areas shall be shown in yellow for current and green for voltage. Each meter shall have a red calibration mark to indicate proper rectified input voltages.

(d) Accuracy. The meters and associated potentiometers and resistors shall permit current and voltage measurement to an accuracy of ± 2% of full scale.

3.5.1.7 Controls. The test set shall have the following controls.

(a) Selector switch. A rotary selector switch located in the lower center of the panel, shall provide meter and circuit switching to accomplish the following:

a. OFF - Disconnect all functions.
b. NO LOAD - Connect meters to battery for voltage measurement only.

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c. ACT. LOAD - Connect activating load and meters to battery for voltage and current measurements.
d. RADIOSONDE - Connect radiosonde and meters to battery for voltage and current measurement.
e. INTERNAL POWER - Connect radiosonde and meters to internal power supply for voltage and current measurements.

(b) Voltage control. The voltage control located on the upper center of the panel shall provide for adjustment of the 115 V. a-c input to the internal power supply. The range of adjustment shall permit the A, B, and C input voltages to the radiosonde to be varied between the limits established for acceptable performance. When the voltage is adjusted to the red calibration mark, the input voltage and current to a standard output load shall be as specified.

3.5 1.8 Internal power supply. The internal power supply shall duplicate as nearly as possible the electrical characteristics of Battery, BA-353/AM.

(a) Regulation. The internal power supply shall be regulated by selection of components and/or design to prevent the output voltages from exceeding full scale deflection of the meters under no load conditions.

(b) Range of acceptable performance. The internal power supply shall provide the following voltages at the currents indicated to operate the radiosonde over its range of acceptable performance.

a.	A Volts	5.2 - 7.5
b	B Volts	90 - 120
c.	C Volts	2.4 - 3.5
d.	A Milliamperes	290 - 350
e.	B Milliamperes	3.0 - 8.0

3.5.1.9 Voltage output and calibration. The internal power supply, meters and calibration resistors shall provide the voltages and/or readings given with the test set in operation as follows:

1. 115 ± 6 volts 60 Hertz a-c applied to connector J103.

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2. 19.5 ± 1% ohms 5 watts standard load resistor connected between terminals A and B of connector J101.

3. 19,000 ± 190 ohms 2 watts standard load resistor connected between terminals B and C of connector J101.

4. Circuit selector switch on internal power position.

5. Calibration button depressed and 115 volts adjust control set so that the A and B current-voltage meter indications are at the red calibration marks.

6. The radiosonde input voltage shall be as follows:

   a. 7.0 ± 0.07 volts d-c between terminals J101-A (-) and J101-B (+) across the 19.5 ohm load.

   b. 114.0 ± 1.0 volts d-c between terminals J101-B (-) and J101-C (+) across the 19,000 ohm load.

7. The A current-voltage meter shall indicate as follows:

   a. A Volts (Normal) 0.34 to 0.36

   b. A Current (Push A Milliamperes) 0.63 to 0.65

8. The B current-voltage meter shall indicate as follows:

   a. B Volts (Normal) 0.44 to 0.46

   b. B Current (Push B Milliamperes) 0.49 to 0.51

   c. C Volts (Push C Volts) 0.74 to 0.76

3.6 Workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any

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of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection of the battery test sets shall be classified as:

(a) First article inspection (see 4.3)

(b) Quality conformance inspection (see 4.4)

(c) Inspection of preparation for delivery (see 4.5)

4.3 First article inspection. First article inspection shall consist of the following:

(a) Contractor's demonstration tests (see 4.3.1)

(b) Service approval tests (see 4.3.2)

4.3.1 Contractor's demonstration tests. First article inspection shall be made on battery test set representative of the production test sets to be supplied under the contract. The government inspector and the procuring activity shall be advised when the inspections are to be conducted so that a representative may be designated to witness or supervise the inspections when so desired. Contractors not having adequate facilities to conduct all required inspections shall obtain the services of a commercial testing laboratory acceptable to the procuring activity. Contractor's demonstration tests shall include the following:

(a) Examination of product (see 4.8.1)

(b) Operational test (see 4.8.2)

(c) Temperature test per MIL-T-21200

(d) Humidity test per MIL-T-21200

(e) Vibration test (see 4.8.3)

(f) Shock test per MIL-T-21200

4.3.1.1 Demonstration test data. The contractor shall submit all data collected in conducting the demonstration tests to the procuring activity for review and approval prior to shipping the equipment to the specified destination for service approval tests.

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4.3.1.2 First article inspection samples. First article inspection samples shall consist of one or more equipments as specified in the contract.

4.3.2 Service approval tests. The service approval model shall consist of one equipment representative of the production equipment to be supplied under the contract. This may be the same equipment that was subjected to the contractor's demonstration tests. This equipment shall be delivered to the destination specified by the Naval Air Systems Command. When the tests of 4.3.1 are supervised, the Naval Air Systems Command may, when conducting service approval tests, omit all tests which will duplicate tests previously conducted by the contractor; however, in conducting service approval tests, the Naval Air Systems Command reserves the right to repeat any test previously conducted when deemed necessary.

4.3.3 First article approval. Approval of the first article model shall be by the Naval Air Systems Command upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the first article model. Prefabrication of production equipment prior to the approval of the first article model is at the contractor's own risk. The approved first article model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The first article model shall not be considered as one of the equipments under the contract, however it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.3.4 Equipment changes. After approval of the first article model, no changes which affect weight, installation interchangeability and interchangeability of maintenance parts or assemblies shall be incorporated in the equipment unless approved by the Naval Air Systems Command. To obtain Naval Air Systems Command approval, the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This shall apply for either a Naval Air Systems Command recommended change or a contractor's proposed change. When required, the contractor shall also prepare an Electronic Material Change (EMC) and Electronic Material Bulletin (EMB) to cover the approved change.

4.4 Quality conformance inspection. The contractor shall furnish all samples and shall be responsible for accomplishing the conformance tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having testing facilities satisfactory to the Naval Air Systems Command shall engage the services of a commercial testing laboratory acceptable to the Naval Air Systems Command. The contractor shall furnish test reports showing quantitative results for all conformance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(a) Individual tests (see 4.4.1)

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(b) Sampling tests (see 4.4.2)

(c) Special tests (see 4.4.3)

(d) Life test (see 4.4.5)

4.4.1 Individual tests. Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, dimensions, tolerances, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests.

(a) Examination of Product (see 4.8.1)

(b) Operational Test (see 4.8.2)

4.4.2 Sampling tests Sampling tests shall be conducted on complete equipments which have successfully passed the individual tests. The following equipments shall be selected by the Government Inspector for sampling tests.

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 10	1
11 to 100	2
101 to 200	1
Over 200	1 for each additional 200 or fraction thereof

4.4.2.1 Scope of tests. As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(a) Complete operational test at ambient room conditions and nominal input voltages, making all necessary measurements to assure that all applicable specification requirements have been met.

(b) Repeat the test of (a) above using the highest and lowest specified input voltages and frequencies.

(c) Reliability test specified in 4.8.4, and conforming with table I, except that the overall test duration shall be 30 hours.

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4.4.3 Special tests. Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling test.

4.4.3.1 Special tests schedule. Selection of equipments for special tests shall be made as follows:

(a) On an early production equipment.

(b) On an equipment midway of the production run.

(c) On an early equipment after an engineering change.

(d) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the Naval Air Systems Command)

Exceptions from the above may be necessary when the number of equipments procured under contract and other factors affecting performance may require a variation to assure satisfactory quality control.

4.4.3.2 Scope of Tests. Special tests shall consist of such tests as approved by the Naval Air Systems Command. Test procedures previously approved for the first article and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the Naval Air Systems Command for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only such tests required to check the characteristics in question need be conducted:

(a) Temperature

(b) Humidity tests

4.4.4 Equipment failure. Should a failure occur during either the sampling or special tests, the following action shall be taken:

(a) Determine the cause of failure.

(b) Determine if the failure is an isolated case or design defect.

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(c) Submit to the Naval Air Systems Command for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occuring in future tests.

(d) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.4.5 Life test. A 300-hour life test shall be conducted at the contractor's plant on an early equipment representative of the current production. The life test shall be performed under the conditions specified in 4.4.5.1. The life test sample shall be selected by the Government Inspector in accordance with the following. (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests.)

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 25	1
26 to 200	1
201 to 500	1
over 500	1 for each additional 500 or fraction thereof

4.4.5.1 Test conditions. The life test shall be conducted under the following simulated service conditions

Temperature	Normal room temperature
Humidity	Room ambient
A-c Voltage	115 ± 6 volts at lowest applicable frequency
D-c Voltage	114 ± 1.0 volts and 7.0 ± 0.07 volts

4.4.5.2 Test periods. The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Temperature stabilization should be in both modes, i.e., off and on, and the equipment shall be turned on and off a minimum of five times and put through each phase of operation.

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4.4.5.3 Performance check. At approximately 8 hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the Naval Air Systems Command.

4.4.5.4 Test data. The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all diode failures. This record shall indicate the following:

(a) Part type number

(b) The circuit reference symbol number

(c) The part function

(d) Name of the manufacturer

(e) Nature of the failure

(f) The number of hours which the part operated prior to failure.

4.4.5.4.1 Failure report. In the event of a failure, the Government Inspector shall be notified immediately. A report shall be submitted to the Naval Air Systems Command upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The Naval Air Systems Command will review such proposals and determine whether they are acceptable.

4.4.5.5 Reconditioning of life test samples. An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all diodes and "wear" items. The "wear" items shall be determined by agreement between the contractor and the Naval Air Systems Command.

(2) After reworking the contractor shall resubmit the equipment for acceptance.

4.5 Inspection of preparation for delivery. Samples from a quantity of shipping containers fully prepared for delivery, just prior to closure, shall be selected at random from each lot in accordance with MIL-STD-105 at Inspection

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Level S-2. The lot size shall be the number of shipping containers in the lot. Each sample selected shall be examined for defects and to determine conformance to all the applicable requirements of Section 5 of this specification. If the number of defective shipping containers exceeds the Acceptable Quality Level of 4.0, the lot represented by the sample shall be rejected.

4.6 Test procedures. The procedures used for conducting first article tests, acceptance tests, and life tests shall be prepared by the contractor and submitted to the Naval Air Systems Command for review and approval. The right is reserved by the Naval Air Systems Command or the Government Inspector to modify the tests or requirements of this specification or the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved procedures are available from previous contracts the Naval Air Systems Command will furnish the procedures to be used for testing.

4.7 Presubmission testing. No item, part, or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.8 Test methods.

4.8.1 Examination of product. Each equipment shall be examined carefully to determine that the material and workmanship requirements of MIL-T-21200 have been met.

4.8.2 Operational Test. Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.8.3 Vibration test. The equipment shall be attached directly to the vibration table or to an intermediate structure which is so designed as to be capable of transmitting the specified magnitudes of vibration to the points of equipment attachment throughout the required test frequency range. The equipment shall be subjected to continuous vibration in each of three mutually perpendicular axes within the frequency range of 10 to 55 cps and having a double amplitude (total excursion) of 0.06 inch. The equipment shall be operating during the vibration test.

4.8.4 Reliability Test Each equipment shall be operated according to the schedule given in Table I. The overall period of operation is six hours during which time the A MILLIAMPERES, B MILLIAMPERES and C VOLTS momentary-contact, push-button switches shall be depressed at the beginning and end of each 90-minute cycled position of the CIRCUIT SELECTOR to check the A and B current indications and the C voltage indications. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be

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permitted which can be made by using such controls and adjustments accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration (10 minutes)	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and an amplitude of ± 3 g's.

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period the equipment shall be operated according to table I and should a failure occur during this 6-hour period, it should be repaired and the test started over, except the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

TABLE I

RELIABILITY TEST PROCEDURE

CIRCUIT SELECTOR Switch Position	Time (min)	INPUT (volts)	OUTPUT LOAD (ohms	METER INDICATIONS				
				NORMALLY A		NORMALLY B		C VOLTS
				volts	current	volts	current	
NO LOAD	90	7 ± 0.07 across P102-1 (+) and P102-4 (-) 114 ± 1 across P102-3 (+) and P102-1 (-)	None	35		45		75
ACT LOAD	90	Same as NO LOAD	None	35	64	45	50	75
RADIOSONDE	90	Same as NO LOAD	19 [illegible] across J101-A and J101-B 19,000 ± 190 across J101-B and J101-C	35	64	.45	50	75
INTERNAL POWER	90	115 ± 6, 60 cycle to J103	Same as RADIOSONDE	.35	64	.45	.50	75

Note - Meter indications have a tolerance of ± 0.01

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5. PREPARATION FOR DELIVERY

5.1 General. All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555.

6. NOTES

6.1 Ordering data. Procurement documents should specify:

6.1.1 Procurement requirements.

(a) Title, number and date of this specification

(b) Level of shipment in accordance with MIL-E-17555

(c) Quantity of samples for first article inspection

6.1.2 Contract data requirements. Items of deliverable data required by this specification are cited in the following paragraphs

Paragraph	Date Requirement	Applicable DID
4.3.1.1	Demonstration test data	
4.4.5.4.1	Failure report	

Such data will be delivered as described on approved DID's (Data Item Description/ DD Form 1664) when specified on DD Form 1423 (Contract Data Requirements List) and incorporated into the applicable contract.

Preparing Activity:
Navy - AS
Project No. 6625-N575

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22-R255

INSTRUCTIONS The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government Fold on lines on reverse side, staple in corner, and send to preparing activity Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements Attach any pertinent data which may be of use in improving this documen If there are additional papers, attach to form and place both in an envelope addressed to preparing activity

DOCUMENT IDENTIFIER AND TITLE

MIL-T-22007A(AS) TEST SET, BATTERY

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO (If "Yes", in what way?)

4 REMARKS

SUBMITTED BY (Printed or typed name and address - Optional)

TELEPHONE NO

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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